

Task5- Phonebook

```
import tkinter as tk
```

```
from tkinter import messagebox
```

```
contacts = []
```

```
# Add Contact
```

```
def add_contact():
```

```
    name = entry_name.get()
```

```
    phone = entry_phone.get()
```

```
    email = entry_email.get()
```

```
    address = entry_address.get()
```

```
    if name == "" or phone == "":
```

```
        messagebox.showwarning("Warning", "Name & Phone required")
```

```
        return
```

```
    contacts.append([name, phone, email, address])
```

```
    listbox.insert(tk.END, f"{name} - {phone}")
```

```
    clear_fields()
```

```
# View / Select Contact
```

```
def show_contact(event):
```

```
    try:
```

```
        index = listbox.curselection()[0]
```

```
        selected = contacts[index]
```

```
        entry_name.delete(0, tk.END)
```

```
        entry_phone.delete(0, tk.END)
```

```
entry_email.delete(0, tk.END)
```

```
entry_address.delete(0, tk.END)
```

```
entry_name.insert(0, selected[0])
```

```
entry_phone.insert(0, selected[1])
```

```
entry_email.insert(0, selected[2])
```

```
entry_address.insert(0, selected[3])
```

```
except:
```

```
    pass
```

```
# Update Contact
```

```
def update_contact():
```

```
    try:
```

```
        index = listbox.curselection()[0]
```

```
        contacts[index] = [
```

```
            entry_name.get(),
```

```
            entry_phone.get(),
```

```
            entry_email.get(),
```

```
            entry_address.get()
```

```
        ]
```

```
        listbox.delete(index)
```

```
        listbox.insert(index, f"{entry_name.get()} - {entry_phone.get()}")
```

```
        clear_fields()
```

```
    except:
```

```
        messagebox.showwarning("Warning", "Select a contact")
```

```
# Delete Contact
```

```
def delete_contact():
```

```
    try:
```

```
        index = listbox.curselection()[0]
```

```

        listbox.delete(index)

        contacts.pop(index)

        clear_fields()

    except:

        messagebox.showwarning("Warning", "Select a contact")


# Search Contact

def search_contact():

    search = entry_search.get().lower()

    listbox.delete(0, tk.END)

    for c in contacts:

        if search in c[0].lower() or search in c[1]:

            listbox.insert(tk.END, f"{c[0]} - {c[1]}")


def clear_fields():

    entry_name.delete(0, tk.END)

    entry_phone.delete(0, tk.END)

    entry_email.delete(0, tk.END)

    entry_address.delete(0, tk.END)


# GUI window

root = tk.Tk()

root.title("Simple Contact Book")

root.geometry("400x500")


# creating Labels & Entries

tk.Label(root, text="Name").pack()

entry_name = tk.Entry(root)

entry_name.pack()

```

```
tk.Label(root, text="Phone").pack()
entry_phone = tk.Entry(root)
entry_phone.pack()
```

```
tk.Label(root, text="Email").pack()
entry_email = tk.Entry(root)
entry_email.pack()
```

```
tk.Label(root, text="Address").pack()
entry_address = tk.Entry(root)
entry_address.pack()
```

```
# create Buttons
```

```
tk.Button(root, text="Add", command=add_contact).pack(pady=5)
tk.Button(root, text="Update", command=update_contact).pack(pady=5)
tk.Button(root, text="Delete", command=delete_contact).pack(pady=5)
```

```
# for Search
```

```
entry_search = tk.Entry(root)
entry_search.pack()
tk.Button(root, text="Search", command=search_contact).pack(pady=5)
```

```
# creating Listbox
```

```
listbox = tk.Listbox(root)
listbox.pack(fill=tk.BOTH, expand=True)
listbox.bind("<<ListboxSelect>>", show_contact)
```

```
root.mainloop()
```

